Trajectory Planning for Mission Completion Time Reduction in UAV-Assisted Data Collection in Wireless Sensor Networks

Bo-Syun Ke
Department of Electrical Engineering
National Cheng Kung University

Unmanned aerial vehicles (UAVs) provide a flexible solution for data collection in wireless sensor networks (WSNs), especially when ground nodes (GNs) are deployed in wide or hard-to-access areas. Mission completion time (MCT) is an important performance metric in UAV-assisted data collection tasks. Existing studies have mainly focused on UAV flight trajectory planning to reduce mission completion time. But the data collection process also has a significant impact on MCT because the transmission rate varies with the distance between the UAV and GNs.

This thesis proposes a trajectory planning method for a single-UAV-assisted WSN data collection task. The proposed method first obtains a shortest geometric reference path that passes through the GN communication regions and uses this path as global directional guidance for local trajectory planning. For each GN, the UAV adjusts its data collection trajectory according to the required data amount. For adjacent GNs with overlapping communication regions, the proposed method further allows service switching within the overlap region to enable a more efficient transition to the next GN. Simulation results show that the proposed method outperforms the representative benchmark methods.

Keywords—UAV, mission completion time, wireless sensor networks, trajectory planning, data collection.

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Chapter 1

Introduction

With the rapid development of the Internet of Things (IoT) and wireless communication technologies, wireless sensor networks (WSNs) have been widely applied in environmental monitoring, smart agriculture, post-disaster detection, military reconnaissance, and other fields [1–3]. In these applications, ground nodes (GNs) are mainly responsible for collecting environmental parameters, such as temperature, humidity, air quality, and water quality. In such IoT applications, each GN usually generates lightweight data packets [4, 5]. However, due to limited battery capacity and hardware cost, GNs are often unable to support long-distance wireless transmission. Therefore, transmitting sensing data back to a data center remains a major challenge.

Unmanned aerial vehicles (UAVs) have become a feasible solution to this problem because of their high mobility, flexible deployment, and ability to establish favorable air-to-ground (A2G) line-of-sight (LoS) communication links [6, 7]. A UAV can serve as an aerial mobile data collector by actively approaching GNs and collecting data within their communication regions. This approach not only reduces the transmission power consumption of GNs and extends the lifetime of sensing devices, but also alleviates communication interruptions caused by terrain blockage [8, 9].

In UAV-assisted data collection tasks, mission completion time (MCT) is a critical performance metric [10]. MCT refers to the total time required for the UAV to depart from the data center, complete data collection from all target GNs, and return to the data center. For time-sensitive IoT applications, such as real-time forest fire monitoring and border intrusion detection, reducing MCT allows the decision center to obtain on-site information more quickly and respond in a timely manner [11, 12]. A shorter MCT also enables the UAV to serve more GNs within its limited battery [13]. Early studies often approximated MCT by the UAV flight time and simplified the problem into a flight distance reduction problem [14]. However, when communication models and data requirements are considered, a shortest geometric path does not necessarily lead to a shorter MCT [15, 16]. According to the Shannon formula, the wireless transmission rate increases with the signal-to-noise ratio (SNR), while the SNR decreases as the communication distance between the UAV and a GN increases [17]. If the UAV passes near the boundary of a GN communication region to reduce flight distance, the flight time can be shortened, but the lower transmission rate can increase the data collection time or hovering time. In contrast, if the UAV moves excessively close to the GN center to improve the transmission rate, additional flight distance and flight time can be introduced. Therefore, reducing MCT requires a proper tradeoff between flight distance and data collection time.

Since the UAV only needs to enter the communication region of each GN to collect data and does not have to fly directly above every GN, this task can be regarded as a path planning problem with neighborhoods, namely the Traveling Salesman Problem with Neighborhoods (TSPN) [18]. However, unlike conventional TSPN, which mainly focuses on geometric path length, UAV-assisted data collection is also affected by communication distance, transmission rate, and data collection time. Therefore, the UAV needs to determine not only the visiting path among communication regions, but also the entry position, exit position, movement depth, and data collection trajectory within each communication region. Existing studies have investigated UAV-assisted WSN data collection from different perspectives. Some studies reduce the geometric path length through GN communication regions [18], while other studies consider flying mode, hovering mode, or V-shaped collection trajectories [19, 20]. These studies show that MCT is affected by both flight time and data collection time. Two issues still require further consideration. First, if global geometric direction information is not considered, local trajectory adjustment can deviate from the overall shorter flight path. Second, when adjacent GN communication regions overlap, restricting service switching to the boundary of a single communication circle can cause redundant boundary transitions and unnecessary back-and-forth movement.

Based on the above observations, this thesis focuses on a single-UAV-assisted WSN data collection task and proposes a trajectory planning method that considers both flight path and data collection efficiency. The proposed method combines global geometric information and a local data collection strategy to reduce the overall MCT of the UAV data collection task. The related mechanisms and implementation details are presented in the following chapters.

The main contributions of this thesis are summarized as follows.

1. A trajectory planning method is developed to reduce the MCT of UAV-assisted WSN data collection by considering both flight trajectory and data collection trajectory.

2. A global-direction-guided local trajectory adjustment method is introduced to keep local service trajectories close to the shortest geometric path and reduce unnecessary flight distance.

3. A flexible service switching strategy is developed for consecutive GNs with overlapping communication ranges, allowing the UAV to use shared coverage regions as switching areas and reduce redundant boundary transitions.

4. Simulation results show that the proposed method reduces the overall MCT compared with the benchmark methods under different data requirements and GN distributions.

The remainder of this thesis is organized as follows. Chapter 2 reviews related studies on trajectory planning for UAV-assisted WSN data collection. Chapter 3 introduces the system model. Chapter 4 describes the proposed trajectory planning method. Chapter 5 presents the simulation setup, and performance evaluation. Finally, Chapter 6 concludes this thesis and discusses possible future research directions.

Chapter 2

Related Work

In UAV-assisted WSN data collection tasks, existing studies can be broadly divided into two categories. The first category focuses on geometric path planning, where the communication range of each GN is regarded as a serviceable region, and the UAV flight path passing through all communication regions is shortened to reduce flight time. The second category considers the data collection process within the communication range, where the UAV adjusts its collection trajectory according to the required data amount and communication conditions. The former provides a shorter geometric reference path, whereas the latter reduces the data collection time.

The study in [18] modeled UAV path planning as a TSPN problem and showed that this problem is NP-hard. In [21] considered a single-UAV-assisted data collection problem by jointly considering trajectory, altitude, velocity, and link scheduling. The study indicated that data transmission can be supported when the UAV is located within the communication range of a ground user, without requiring the UAV to fly directly above the ground user. The original mission time problem was transformed into a trajectory length problem. This method mainly focuses on shortening the geometric path length.

Reference [15] indicated that simply shortening the flight distance can move the UAV farther away from sensor nodes, thereby reducing the transmission rate and increasing the time required to complete data reception. The results show that the flight distance, flight speed, and communication rate in UAV data collection tasks are not independent and should be considered jointly. The effect of the collection point location on mission time was further discussed in [16], where it was shown that the UAV data collection point should not be determined solely by the shortest flight distance or the best communication quality. If the UAV flies directly above a node, better communication quality can be achieved, but the flight distance can increase. In contrast, if the UAV collects data near the boundary of the communication range, the flight path can be shortened, but the longer distance between the UAV and the node reduces the transmission rate and increases the data collection time. This observation suggests that the collection point location should be determined by jointly considering the flight cost and the communication condition.

In [19], FM and HM were introduced as two data collection modes for UAVs. In FM, the UAV can continuously receive data while flying, which is suitable for nodes with smaller data requirements or better communication conditions. In HM, the UAV hovers at a specific location for data collection, which is suitable for nodes with larger data requirements or poorer communication conditions. The distinction between FM and HM allows the UAV to adjust its collection mode according to the node status and data requirement, instead of serving all nodes with a fixed collection mode.

For user scheduling and UAV trajectory design, [22] investigated the completion time problem for a single UAV collecting data from multiple sensor nodes. It was proved that, under specific conditions, a suitable UAV trajectory has an SHF structure, in which the UAV hovers at selected locations for data collection and flies at the maximum speed between these hovering locations. The authors also emphasized the importance of segment-based scheduling, meaning that the UAV mainly serves a specific sensor during a particular flying or hovering segment instead of frequently switching among different sensors.

In [12] further investigated the impact of the data collection trajectory on the MCT and proposed a V-shaped collection trajectory. If the UAV flies into the interior of the communication range, part of the flight distance increases, but the SNR and transmission rate can be improved, thereby reducing the data collection time and the overall mission time. The work also pointed out that the service trajectories of adjacent nodes are coupled. The previous node affects the starting position when the UAV serves the next node. Therefore, if the service trajectory of each node is determined independently, additional flight distance can be introduced for subsequent nodes. In other words, locally reducing the data collection time of one node does not necessarily reduce the overall MCT. The UAV still needs to consider the continuous flight direction and service connection between adjacent nodes.

The above studies show that trajectory planning for UAV-assisted data collection mainly involves two interrelated design aspects. First, geometric path planning methods reduce the flight distance for the UAV to pass through multiple communication regions [18, 21]. Second, data collection trajectory design indicates that the UAV can moderately move into the communication range according to the required data amount and communication conditions, thereby improving the transmission rate and reducing the data collection time [12, 16]. In addition, existing studies have explained the close relationship between UAV flight behavior and communication conditions from the perspectives of flight speed and transmit power control, as well as FM/HM mode switching [15, 19, 22].

Based on the above research context, this thesis focuses on a single-UAV-assisted WSN data collection task and investigates how to jointly consider the flight trajectory, the data collection trajectory within the communication range, and the service connection between adjacent nodes. Existing studies have shown the importance of geometric path shortening and data collection within the communication range. If trajectory planning mainly relies on geometric path length, the effect of the distance variation between the UAV and the GN on the data collection time cannot be fully reflected. In contrast, if the local data collection trajectory lacks a global geometric direction reference, the trajectory can gradually deviate from the shorter overall flight direction. In addition, when the communication ranges of adjacent or consecutive nodes overlap, the overlapping region provides additional flexibility for service switching. Effectively using this geometric property can help reduce unnecessary boundary switching and back-and-forth flight. Therefore, this study focuses on maintaining a shorter overall flight distance while considering the data collection efficiency within the communication range, and further utilizes the overlapping regions between adjacent communication ranges to improve service switching flexibility, thereby reducing the overall MCT.

Chapter 3

System Model

3.1 System Model

As shown in Fig. 3.1, this thesis considers a UAV-assisted wireless sensor network in which a single UAV is employed to collect data from multiple ground nodes (GNs). The UAV departs from the data center, visits the communication regions of the GNs to collect their data, and returns to the data center after completing the data collection task. Let $\mathcal{N} = \{g_1, g_2, \ldots, g_n\}$ denote the set of GNs, where n is the number of GNs. A three-dimensional Cartesian coordinate system is adopted, where the GNs are located on the ground plane and the UAV flies at a fixed altitude H. The elevation angle between the UAV and a GN is denoted by θ, and the maximum flying speed of the UAV is denoted by $v_{\max}$.

During the data collection process, the UAV may operate in flying mode (FM) or hovering mode (HM). In FM, the UAV collects data while moving within the communication region of a GN. In HM, if the data collected along the flying trajectory is insufficient, the UAV hovers directly above the corresponding GN for additional data collection.

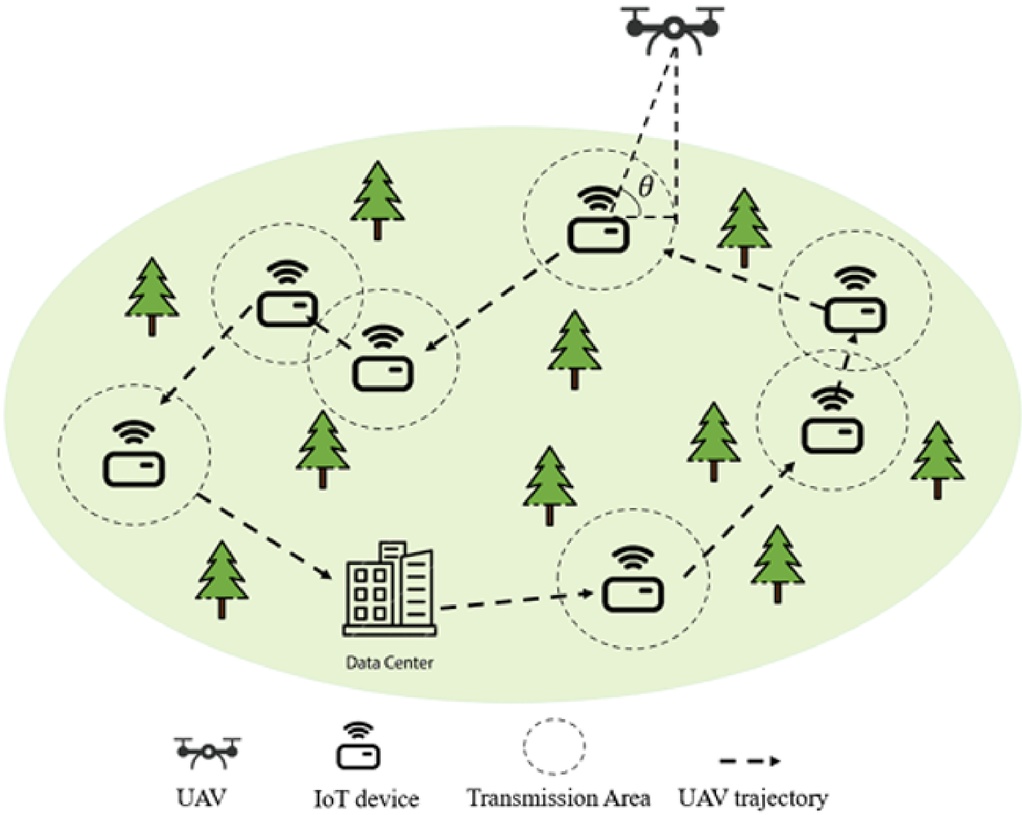

Figure 3.1: Illustration of the UAV data collection process.

1) Transmission Model: A delay-tolerant UAV-assisted WSN is considered, in which each ground node g_i stores sensing data and uploads the data to the UAV when the UAV enters its communication region. Each ground node is equipped with a single omnidirectional antenna and transmits with a fixed power P_t over a channel bandwidth B. The uplink transmission rate from GN g_i to the UAV at time t is expressed as

$$R_i(t) = B \log_2 \left(1 + \gamma_i(t)\right) \tag{3.1}$$

in which $\gamma_i(t)$ denotes the received signal-to-noise ratio (SNR) from GN g_i to the UAV at time t. The SNR is given by

$$\gamma_i(t) = \frac{P_t}{\sigma^2 L_{dB_i}(t)} \tag{3.2}$$

where σ^2 denotes the noise power at the UAV receiver, and $L_{dB_i}(t)$ represents the air-to-ground path loss between GN g_i and the UAV at time t. The UAV can successfully collect data only when the received SNR satisfies $\gamma_i(t) \geq \bar{\gamma}$. To avoid transmission interference among ground nodes, only the ground node being served is allowed to transmit data to the UAV during its corresponding service interval.

2) Channel Model: Considering that the mission completion time generally exceeds the channel coherence time and that the UAV position changes during the mission, a statistical channel model is adopted. The path loss from GN g_i to UAV can be expressed as

$$L_{dB_i}(t) = \left(\frac{4\pi f_c}{c}\right)^2 l_i^{\eta} \left[P_{\mathrm{LoS}_i}(t) \left(\xi_{\mathrm{LoS}} - \xi_{\mathrm{NLoS}}\right) + \xi_{\mathrm{NLoS}}\right] \tag{3.3}$$

where f_c denotes the carrier frequency, c is the speed of light, and η is the path-loss exponent. The parameters ξ_{LoS} and ξ_{NLoS} represent the average additional path losses under LoS and NLoS propagation conditions, respectively. In addition, $l_i(t)$ denotes the three-dimensional distance between GN g_i and the UAV, which is given by

$$l_i(t) = \sqrt{\left(U_x(t) - GN_{ix}\right)^2 + \left(U_y(t) - GN_{iy}\right)^2 + H^2} \tag{3.4}$$

where (GN_{ix}, GN_{iy}) is the horizontal coordinate of GN g_i, $(U_x(t), U_y(t))$ is the horizontal coordinate of the UAV at time t, and H is the fixed flying altitude of the UAV.

The LoS probability between GN g_i and the UAV is modeled as

$$P_{\mathrm{LoS}_i}(t) = \frac{1}{1 + \alpha \exp\left[-\beta\left(\theta_i(t) - \alpha\right)\right]} \tag{3.5}$$

where α and β are parameters determined by the propagation environment. The elevation angle from GN g_i to the UAV at time t is denoted by $\theta_i(t)$, which is defined as

$$\theta_i(t) = \frac{180}{\pi} \arcsin\left(\frac{H}{l_i(t)}\right). \tag{3.6}$$

According to this channel model, the path loss is affected by both the UAV–GN distance and the elevation angle. When the UAV moves closer to a GN, the three-dimensional distance decreases and the elevation angle increases, which generally improves the LoS probability and reduces the average path loss. Therefore, the UAV trajectory directly affects the received SNR and the corresponding data collection time.

3.2 Problem Formulation

In the considered single-UAV data collection mission, all GNs must be served within one flight. The UAV starts from the data center, collects the required data from each GN, and returns to the data center. For a given transmit power P_t, the communication radius of each GN is determined by the SNR threshold $\bar{\gamma}$. Let D denote the maximum horizontal distance between the UAV and a GN such that the received SNR is equal to $\bar{\gamma}$. When the UAV is located inside the communication region with radius D, the received SNR is no less than the required threshold, and the UAV can successfully receive data from the corresponding GN. The value of D is obtained according to the adopted LoS-based air-to-ground channel model.

The MCT of the UAV is obtained by accumulating the service time required for all GNs. For each GN, the service time consists of the flight time required to reach its communication region and the data collection time spent within that region. The index i follows the service sequence along the UAV trajectory. The GN visiting sequence is first determined by a genetic algorithm (GA), and the proposed method plans the complete UAV trajectory based on this sequence. As shown in Fig. 3.2, GN g_i is served independently, whereas GN g_{i+1} and GN g_{i+2} share an overlapping communication region.

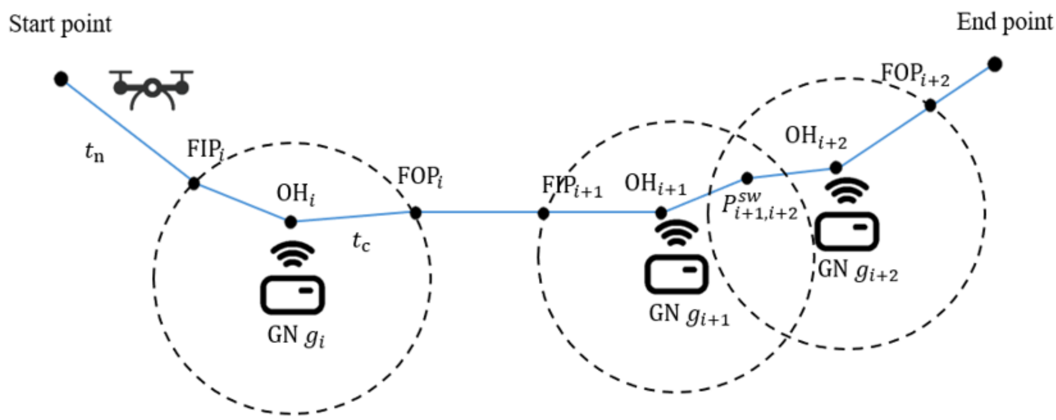

Figure 3.2: UAV data collection process.

UAV service trajectory is described by three key points: the flying-in point FIP_i, the optimal hovering or turning point OH_i, and the flying-out point FOP_i. The UAV first flies from the previous service point to FIP_i, and this flight duration is denoted by t_{n_i}. After entering the communication region of GN g_i, the UAV collects data while moving from FIP_i to FOP_i through OH_i. The corresponding data collection time is denoted by t_{c_i}.

When the communication regions of two consecutive GNs do not overlap, the UAV enters and leaves the communication region of each GN through its own FIP and FOP, respectively. In this case, FIP_i and FOP_i are located on the boundary of the communication region with radius D, and the data collection trajectory of GN g_i is represented as

$$\mathrm{FIP}_i \rightarrow \mathrm{OH}_i \rightarrow \mathrm{FOP}_i. \tag{3.7}$$

When the communication regions of GN g_{i+1} and GN g_{i+2} overlap, the service transition between the two GNs is represented by a common point selected from the overlapping region. Therefore, the FOP of GN g_{i+1} and the FIP of GN g_{i+2} are not forced to lie on the boundaries of their respective communication regions. To describe this service transition, this thesis introduces an overlap switching point, denoted by $P^{\mathrm{sw}}_{i+1,i+2}$. The switching point satisfies

$$P^{\mathrm{sw}}_{i+1,i+2} \in D_{i+1} \cap D_{i+2} \tag{3.8}$$

where D_i denotes the communication region of GN g_i. The switching point simultaneously serves as the FOP of GN g_{i+1} and the FIP of GN g_{i+2}, which is expressed as

$$\mathrm{FOP}_{i+1} = P^{\mathrm{sw}}_{i+1,i+2} = \mathrm{FIP}_{i+2}. \tag{3.9}$$

The service trajectory across the overlapping communication regions of GN g_{i+1} and GN g_{i+2} can be represented as

$$\mathrm{FIP}_{i+1} \to \mathrm{OH}_{i+1} \to P^{\mathrm{sw}}_{i+1,i+2} \to \mathrm{OH}_{i+2} \to \mathrm{FOP}_{i+2}. \tag{3.10}$$

This representation indicates that the UAV completes the data collection of GN g_{i+1} before reaching the switching point, and then continues the data collection of GN g_{i+2} after passing through the same point. This setting keeps the trajectory continuous and allows the UAV to complete the service transition within the overlapping communication region, thereby avoiding unnecessary boundary departure and re-entry between consecutive GNs. For a group of consecutive overlapping GNs, the same concept is applied by defining an overlap switching point between each pair of adjacent GNs in the group.

Except for hovering-based data collection, the UAV flies at the maximum speed $v_{\max}$. Therefore, the flight time is proportional to the corresponding flight distance. For the flight segment from the previous service point to FIP_i, the flight time t_{n_i} is expressed as

$$t_{n_i} = \frac{d_{i1}}{v_{\max}} \tag{3.11}$$

where

$$d_{i1} = \left\| \mathrm{FIP}_i - \mathrm{FOP}_{i-1} \right\|. \tag{3.12}$$

For $i = 1$, FOP_0 denotes the data center. If GN g_i and the previously served GN overlap, FIP_i can be the corresponding overlap switching point, and the fly-in distance is adjusted accordingly. After completing the data collection task of the last GN, the UAV returns to the data center. The return time is given by

$$t^r = \frac{\|\mathrm{DC} - \mathrm{FOP}_N\|}{v_{\max}} \tag{3.13}$$

where DC denotes the data center.

The data collection time of GN g_i, denoted by t_{c_i}, is evaluated according to the adopted V-shaped collection trajectory. During this period, the UAV collects data while flying within the communication region. If the data collected during flight is insufficient, the UAV hovers directly above the corresponding GN to complete the remaining data collection. Let Q_i denote the required data amount of GN g_i. The accumulated received data during the collection interval must satisfy

$$\int_0^{t_{c_i}} R_i(t)\, dt \geq Q_i. \tag{3.14}$$

The MCT of the UAV is expressed as

$$T = \sum_{i=1}^{N} (t_{c_i} + t_{n_i}) + t^r. \tag{3.15}$$

Let $\mathcal{P}$ denote the set of service points associated with the UAV trajectory, including the FIPs, OHs, FOPs, and the overlap switching points when communication regions overlap. Based on the above definitions, the overall MCT reduction problem is formulated as

$$
\begin{aligned}
\textbf{(P1):} \quad & \min_{U(t),\mathcal{P}} \left[\sum_{i=1}^{N} (t_{c_i} + t_{n_i}) + t^r \right] \\
\text{s.t.} \quad & \gamma_i(t) \geq \bar{\gamma} \\
& \mathrm{FIP}_i, \mathrm{OH}_i, \mathrm{FOP}_i \in D_i \\
& P^{\mathrm{sw}}_{i,i+1} \in D_i \cap D_{i+1},\ i \in \mathcal{O} \\
& \int_0^{t_{c_i}} R_i(t)\, dt \geq Q_i
\end{aligned}
\tag{3.16}
$$

Here, $\mathcal{O}$ denotes the set of consecutive GN indices whose communication regions overlap, i.e., $D_i \cap D_{i+1} \neq \emptyset$. Constraint (17a) ensures that the received SNR satisfies the required threshold during data collection. Constraint (17b) requires the key service points of each GN to lie within the corresponding communication region. Constraint (17c) defines the feasible region of the overlap switching point. Constraint (17d) guarantees that the required data amount of each GN is collected.

Problem **P1** is difficult to solve directly because the flight trajectory and the data collection trajectory are coupled. The selection of service points in $\mathcal{P}$ affects both the flight time t_{n_i} and the data collection time t_{c_i}. Moreover, when consecutive communication regions overlap, the overlap switching points affect the continuity of the service trajectory across adjacent GNs. Therefore, this thesis addresses **P1** through two trajectory planning components in the proposed solution: single-node trajectory planning and overlapping-node group trajectory planning. The RPA mechanism provides reference-path attractors for both components, whereas the BO strategy is specifically applied to overlapping-node groups to determine feasible service switching points within overlapping communication regions.

Chapter 4

Proposed Solution

This chapter presents the proposed trajectory planning method for reducing the mission completion time (MCT) in UAV-assisted WSNs. Since the MCT is determined by both flight time and data collection time, a trajectory with the shortest geometric length does not necessarily lead to the shortest mission duration. Therefore, the proposed method is developed based on two design considerations: reducing both the flight distance and the data collection time of each GN.

To achieve this objective, this thesis first constructs a shortest geometric reference path to provide global directional information. Based on this reference path, the Reference-Path Attractor (RPA) mechanism is used to guide the subsequent local trajectory planning process. The final UAV trajectory is then determined according to the data collection requirement of each GN and the spatial relationship among communication regions. Through this design, the UAV trajectory can remain close to the shortest geometric reference path while allowing the UAV to move deeper into the communication region when such movement can improve the transmission rate and reduce the data collection time.

Based on the spatial relationship among GNs, the proposed trajectory planning process is divided into two cases. For an independent GN whose communication region does not overlap with the communication regions of other GNs, this thesis applies single-node trajectory planning to determine the corresponding service trajectory. For consecutive GNs with overlapping communication regions, this thesis adopts the Boundary-Overlap Strategy (BO), which treats the overlapping GNs as a group and uses the flexibility provided by the overlapping regions to determine the service trajectory.

The remainder of this chapter is organized as follows. Section 5.1 introduces the construction of the shortest geometric reference path and the Reference-Path Attractor (RPA) mechanism. Section 5.2 presents the single-node trajectory planning method. Section 5.3 describes the overlapping-node group trajectory planning method based on BO.

4.1 Reference Path Construction and RPA Mechanism

The purpose of constructing a shortest geometric reference path is to provide a global geometric direction for the subsequent trajectory planning process. If the trajectory segment associated with each GN is determined only according to the data collection time of the current GN, the selection of FIP and FOP may lack guidance from the shortest geometric reference path. As a result, the subsequent trajectory may deviate from the shortest geometric reference path, leading to an increase in flight distance. Therefore, this thesis first computes a shortest geometric reference path that passes through all GN communication regions and then extracts reference-path attractors from this path. In this thesis, an attractor is defined as a reference point extracted from the shortest geometric reference path within the communication region of a GN. The attractor is used as a directional reference when evaluating candidate FIP/FOP pairs in later trajectory planning, but the UAV is not required to pass through the attractor.

Following the service order defined in the problem formulation, the index i denotes the i-th GN visited by the UAV, and the communication region of this GN is denoted by $\mathcal{D}_i$, as defined in the system model. For the given visiting sequence, this thesis computes a shortest geometric path that passes through all communication regions $\mathcal{D}_i$. To formulate the reference path construction problem, two auxiliary reference-path points are introduced for each communication region. Let $\mathbf{r}_i^{\text{in}}$ and $\mathbf{r}_i^{\text{out}}$ denote the reference-path entry point and reference-path exit point associated with $\mathcal{D}_i$, respectively. These two points are used only for constructing the shortest geometric reference path and are not the final FIP and FOP used for data collection. The shortest geometric reference path is obtained by solving

$$\begin{aligned}
\min_{\{\mathbf{r}_i^{\text{in}},\mathbf{r}_i^{\text{out}}\}_{i=1}^N} \quad & \left\|\mathbf{r}_1^{\text{in}} - \mathbf{q}_0\right\| + \sum_{i=1}^{N} \left\|\mathbf{r}_i^{\text{out}} - \mathbf{r}_i^{\text{in}}\right\| \\
& + \sum_{i=1}^{N-1} \left\|\mathbf{r}_{i+1}^{\text{in}} - \mathbf{r}_i^{\text{out}}\right\| + \left\|\mathbf{q}_0 - \mathbf{r}_N^{\text{out}}\right\| \\
\text{s.t.} \quad & \mathbf{r}_i^{\text{in}} \in \mathcal{D}_i \\
& \mathbf{r}_i^{\text{out}} \in \mathcal{D}_i.
\end{aligned} \tag{4.1}$$

where $\mathbf{q}_0$ denotes the data center. The first term represents the distance from the data center to the reference-path entry point of the first visited GN. The second term represents the geometric segment inside each communication region. The third term represents the distance from the reference-path exit point of the current GN to the reference-path entry point of the next visited GN. The last term represents the return distance from the last communication region to the data center. Since the objective function is composed of Euclidean norm terms and the feasible communication regions are convex, (5.1) can be formulated as a SOCP problem.

The segment from $\mathbf{r}_i^{\text{in}}$ to $\mathbf{r}_i^{\text{out}}$ represents only the portion of the shortest geometric reference path inside the communication region $\mathcal{D}_i$, rather than the actual data collection trajectory of the i-th GN. The final FIP, OH, and FOP are determined later according to the data requirement of the GN and the corresponding data collection time. Therefore, the shortest geometric reference path provides geometric reference information for the subsequent trajectory planning process.

After the shortest geometric reference path is obtained, the RPA mechanism extracts one attractor for each GN. Let the shortest geometric reference path be denoted by $\mathcal{P}_{\text{ref}}$. The path $\mathcal{P}_{\text{ref}}$ is scanned according to the traveling direction from the data center. For the i-th GN, the candidate attractor points are the points on $\mathcal{P}_{\text{ref}}$ that lie within the communication region $\mathcal{D}_i$. Let $\ell(\mathbf{p})$ denote the cumulative path length from $\mathbf{q}_0$ to point $\mathbf{p}$ along $\mathcal{P}_{\text{ref}}$. The attractor of the i-th GN is defined as

$$\mathbf{a}_i = \arg\min_{\mathbf{p}\in\mathcal{P}_{\text{ref}}\cap\mathcal{D}_i} \ell(\mathbf{p}). \tag{4.2}$$

As shown in Fig. 4.1, the shortest geometric path passes through the communication regions of the GNs according to the visiting direction. The triangular markers indicate the reference-path attractors extracted from this path. Each attractor is selected from the portion of the shortest geometric path that lies within the corresponding communication region.

The attractor $\mathbf{a}_i$ is selected as the first reference-path point that lies within the communication region of the i-th GN along the traveling direction. This early-entry selection preserves a longer available data collection interval along the reference path.

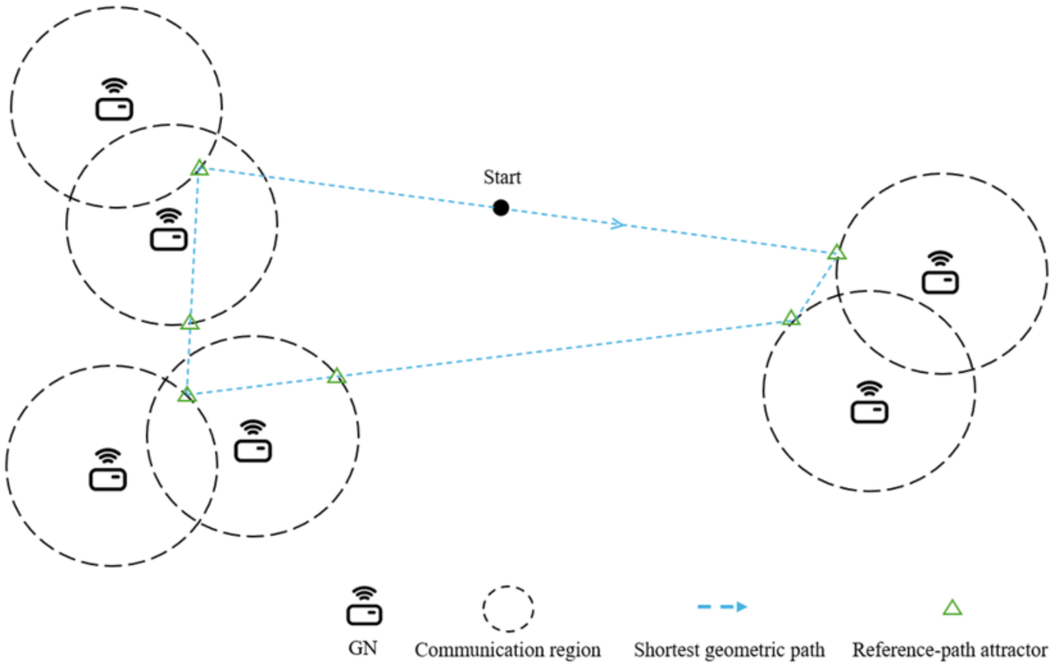

Figure 4.1: Illustration of the reference-path attractors.

Therefore, $\mathbf{a}_i$ is used as the reference-path entry point associated with the i-th GN. The terminal attractor is set as the data center, $\mathbf{a}_{N+1} = \mathbf{q}_0$.

The influence of the attractor can be explained by decomposing the flight-time term associated with the i-th GN. Let $\mathbf{P}_i^{\text{pre}}$ denote the point immediately before serving the i-th GN. The point $\mathbf{P}_i^{\text{pre}}$ can be the FOP of the previous GN, an overlap switching point, or the data center when $i = 1$. For a candidate FIP_i, the fly-in time is expressed as

$$t_{n_i}^{\text{in}} = \frac{\|\text{FIP}_i - \mathbf{P}_i^{\text{pre}}\|}{v_{\max}}. \tag{4.3}$$

After the data collection process of the i-th GN is completed, the outgoing flight direction is evaluated with respect to the next attractor $\mathbf{a}_{i+1}$. The RPA-guided fly-out time is expressed as

$$t_{n_i}^{\text{out}} = \frac{\|\mathbf{a}_{i+1} - \text{FOP}_i\|}{v_{\max}}. \tag{4.4}$$

Accordingly, the flight-time term considered in the RPA-guided trajectory planning process is written as

$$t_{n_i}^{\mathrm{RPA}} = t_{n_i}^{\mathrm{in}} + t_{n_i}^{\mathrm{out}}. \tag{4.5}$$

The fly-out term $t_{n_i}^{\mathrm{out}}$ shows how the next attractor affects the evaluation of the current FOP. When different FIP/FOP pairs are considered in the trajectory evaluation process, the selected FOP determines the flight time toward the next attractor. Therefore, the RPA mechanism affects the selection of the FIP/FOP pair through the RPA-guided flight-time term.

A candidate FIP/FOP pair is evaluated not only according to the data collection time of the current GN, but also according to the flight time toward the reference direction derived from the shortest geometric reference path. The UAV trajectory is not constrained to pass through the attractor. The complete candidate evaluation rule is described in Section 5.2.

Algorithm 1 summarizes the procedure for constructing the shortest geometric reference path and extracting the corresponding reference-path attractors. Based on the extracted attractors and the RPA-guided flight-time term, Section 5.2 presents the single-node trajectory planning method, including the generation of candidate FIP/FOP pairs, the determination of OH, and the evaluation of t_{c_i}. Section 5.3 further describes how the trajectory planning process is modified when consecutive GNs have overlapping communication regions.

Algorithm 1: Reference Path Construction and Attractor Extraction

Input: Communication regions $\{\mathcal{D}_i\}_{i=1}^{N}$ in the visiting order; data center $\mathbf{q}_0$

Output: Shortest geometric reference path $\mathcal{P}_{\mathrm{ref}}$; reference-path attractors $\{\mathbf{a}_i\}_{i=1}^{N+1}$

```
Solve the SOCP problem in (??) to obtain {r_i^in, r_i^out}_{i=1}^N;
Construct P_ref by connecting q_0, r_1^in, r_1^out, ..., r_N^in, r_N^out, and q_0 in sequence;
Define ℓ(p) as the cumulative path length from q_0 to point p along P_ref;
for i = 1 to N do
    Extract the first reference-path point that lies within D_i;
    Set a_i ← arg min_{p ∈ P_ref ∩ D_i} ℓ(p);
Set the terminal attractor as a_{N+1} ← q_0;
return P_ref and {a_i}_{i=1}^{N+1};
```

4.2 Single-Node Trajectory Planning

This section describes the trajectory planning process for an independent GN. An independent GN is defined as a GN whose communication region does not overlap with the communication regions of other GNs. Based on the RPA-guided flight-time term defined in Section 5.1, this planning process selects a suitable FIP/FOP pair while considering the data collection time inside the communication region. The corresponding OH is determined during the evaluation of the V-shaped data collection trajectory.

As shown in Fig. 4.2, the UAV collects data while moving inside the communication region of the GN. Candidate FIP and FOP points are generated on the communication boundary of the GN. Let $\mathcal{F}_i^{\mathrm{in}}$ and $\mathcal{F}_i^{\mathrm{out}}$ denote the candidate sets of FIP and FOP, respectively. For each candidate pair $(\mathrm{FIP}_i, \mathrm{FOP}_i)$, the corresponding OH_i and the data collection time t_{c_i} are evaluated according to the required data amount C_i.

For each candidate pair, the corresponding V-shaped trajectory is evaluated to determine whether the required data amount can be collected during flight. If the flying collection is sufficient, FM is adopted. If the flying collection is insufficient, HM is

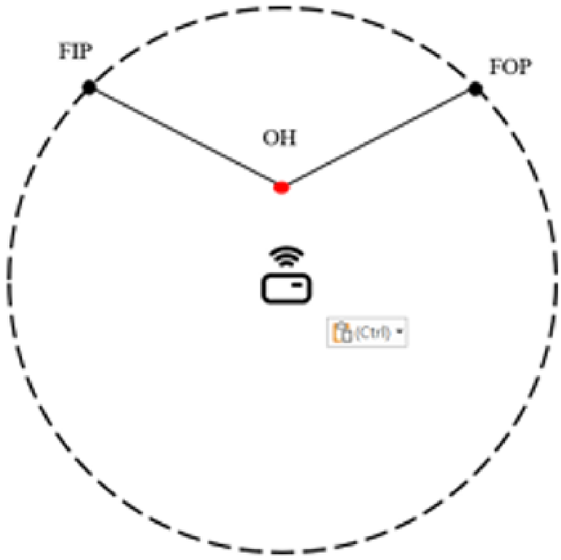

Figure 4.2: V-shaped trajectory for single-node data collection.

adopted, and the UAV collects the remaining data by hovering. After the data collection time is obtained for all candidate pairs, the candidate pair is selected by considering both t_{c_i} and the RPA-guided flight-time term $t_{n_i}^{\mathrm{RPA}}$.

For a given FIP/FOP pair, the flying distance inside the communication region is determined by the sum of the distance from FIP to OH and the distance from OH to FOP. According to the geometric property of an ellipse, all points with the same sum of distances to FIP_i and FOP_i form an elliptical curve. Therefore, different V-shaped trajectories can be evaluated by adjusting OH_i and the corresponding internal flying distance. When the required data amount can be collected during flight, a bisection search is used to determine a feasible V-shaped flying trajectory that satisfies the data collection requirement.

The amount of data collected during flight is calculated by integrating the instantaneous transmission rate along the trajectory:

$$C_i^{\mathrm{FM}} = \int_0^{t_i^{\mathrm{FM}}} R_i(t)\, dt. \tag{4.6}$$

where t_i^{FM} denotes the flying collection time of the candidate V-shaped trajectory. If $C_i^{\mathrm{FM}} \geq C_i$, the required data amount can be collected in FM, and the data collection time is given by $t_{c_i} = t_i^{\mathrm{FM}}$. If $C_i^{\mathrm{FM}} < C_i$, the flying trajectory cannot collect all required data. The UAV then operates in HM. In HM, OH is used as the hovering point, and the UAV collects the remaining data at this point. The additional hovering time is calculated as

$$t_i^{\mathrm{H}} = \frac{C_i - C_i^{\mathrm{FM}}}{R_i(\mathrm{OH}_i)}. \tag{4.7}$$

Therefore, the data collection time in HM is

$$t_{c_i} = t_i^{\mathrm{FM}} + t_i^{\mathrm{H}}. \tag{4.8}$$

After t_{c_i} is obtained for all candidate FIP/FOP pairs, each candidate is evaluated together with the RPA-guided flight-time term. The selected pair is determined by

$$\min_{\mathrm{FIP}_i, \mathrm{FOP}_i} \; t_{c_i}\left(\mathrm{FIP}_i, \mathrm{OH}_i, \mathrm{FOP}_i\right) + t_{n_i}^{\mathrm{RPA}}. \tag{4.9}$$

In (5.12), OH_i is determined during the evaluation of t_{c_i}. Therefore, the selected pair is determined by considering both the data collection time inside the communication region and the RPA-guided flight time. After the trajectory of the current GN is determined, the resulting FOP is used as the starting point for planning the next GN. When consecutive GNs have overlapping communication regions, the corresponding overlapping-node trajectory planning method is described in Section 5.3.

4.3 Overlapping-Node Group Trajectory Planning

This section describes the trajectory planning method for GNs with overlapping communication regions. For an independent GN, the FIP and FOP candidates are generated on the communication boundary of a single GN, as described in Section 5.2.

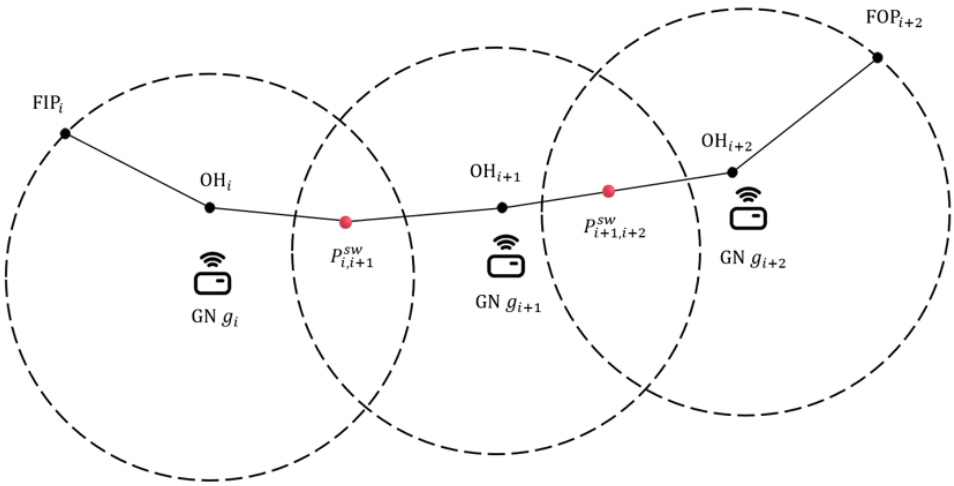

Figure 4.3: Boundary-Overlap Strategy for overlapping GNs.

When the communication regions of adjacent GNs overlap, the overlapping region provides additional flexibility for service transition. If the UAV is still forced to leave one communication region from its boundary and enter the next communication region from another boundary point, unnecessary flight distance may be introduced.

To use the overlapping region between adjacent GNs, this thesis proposes the Boundary-Overlap Strategy (BO). BO allows the switching point $P^{\mathrm{sw}}_{i,i+1}$, defined in the system model, to be selected from the overlapping region of g_i and g_{i+1}. In this way, the UAV can connect the service trajectory of two adjacent GNs through the overlapping region, instead of performing an additional boundary-to-boundary transition.

As shown in Fig. 4.3, multiple adjacent GNs with overlapping communication regions are considered as one group. Let

$$G_m = \{g_i, g_{i+1}, \ldots, g_j\} \tag{4.10}$$

denote the m-th overlapping-node group, where g_i and g_j are the first and last GNs in the group, respectively. Here, G_m denotes a set of GNs, while g_i denotes an individual GN. For any two adjacent GNs g_k and g_{k+1} in G_m, where $k = i, \ldots, j-1$, their communication regions overlap. The UAV enters the group from the FIP of g_i, passes through the switching points in the overlapping regions and leaves the group from the FOP of g_j. For example, if the group contains g_i, g_{i+1}, and g_{i+2}, the service trajectory inside the group can be represented as

$$\mathrm{FIP}_i \rightarrow \mathrm{OH}_i \rightarrow P^{\mathrm{sw}}_{i,i+1} \rightarrow \mathrm{OH}_{i+1} \rightarrow P^{\mathrm{sw}}_{i+1,i+2} \rightarrow \mathrm{OH}_{i+2} \rightarrow \mathrm{FOP}_{i+2}. \tag{4.11}$$

In this sequence, $\mathbf{P}^{\mathrm{sw}}_{i,i+1}$ is used as the exit point of g_i and the entry point of g_{i+1}. Similarly, $\mathbf{P}^{\mathrm{sw}}_{i+1,i+2}$ connects the service trajectories of g_{i+1} and g_{i+2}. Therefore, the group is planned as a single service unit with one entry point and one exit point.

For each overlapping region, BO selects the switching point from a finite candidate set instead of searching over the entire continuous overlapping region. The candidate set is generated using two representative sampling directions. The first direction follows the line segment connecting the centers of the two adjacent GNs. Since different adjacent GN pairs may have different overlap depths. Therefore BO can adjust whether the service transition should occur earlier or later within the overlapping region. The second direction follows a perpendicular sampling line inside the overlapping region. This direction provides candidates with different inward depths within the overlapping region. Since different data amounts may require different collection depths, the perpendicular sampling line allows BO to adjust the switching location according to the data collection requirement. By combining these two representative directions, the T-shaped sampling method generates switching candidates for the overlapping region without searching all possible points in the continuous region.

For the m-th overlapping-node group G_m, a candidate point sequence is written as

$$\mathcal{S}_m = \left[\mathrm{FIP}_i, \mathbf{P}^{\mathrm{sw}}_{i,i+1}, \ldots, \mathbf{P}^{\mathrm{sw}}_{j-1,j}, \mathrm{FOP}_j\right]. \tag{4.12}$$

The first point in $\mathcal{S}_m$ is the group entry point, and the last point is the group exit point. The intermediate points are switching points selected from the overlapping regions Given a candidate sequence, each GN uses the previous point in the sequence as its entry point and the next point as its exit point. After a candidate sequence $\mathcal{S}_m$ is determined, the data collection time of each GN in the group is evaluated according to the V-shaped trajectory model described in Section 5.2. The collection-time term of the group is defined as the sum of the data collection times of all GNs in the group:

$$t_{c,m} = \sum_{k=i}^{j} t_{c_k} \tag{4.13}$$

In addition to the collection-time term, the flight time before entering and after leaving the group must also be considered. Since the overlapping-node group is served as one continuous trajectory segment, the group has one entry point and one exit point. The flight-time term of the group is expressed as

$$t_{n,m} = \frac{\|\mathrm{FIP}_i - \mathrm{FOP}_{i-1}\|}{v_{\max}} + \frac{\|\mathbf{a}_{j+1} - \mathrm{FOP}_j\|}{v_{\max}}. \tag{4.14}$$

Therefore, each candidate sequence is evaluated by

$$T_m = t_{c,m} + t_{n,m}. \tag{4.15}$$

In (5.18), T_m denotes the evaluation time of a candidate sequence for the m-th overlapping-node group. The term $t_{c,m}$ represents the total data collection time of the GNs in the group, while $t_{n,m}$ represents the flight time associated with entering and leaving the group. The candidate sequence with the smallest T_m is selected as the service trajectory of the group. In (5.17), the first term represents the flight time from the FOP of the previous GN to the entry point of the group, and the second term represents the flight time from the exit point of the group toward the next attractor.

Algorithm 2: Boundary-Overlap Strategy for an Overlapping-Node Group

Input: Overlapping-node group $G_m = \{g_i, g_{i+1}, \ldots, g_j\}$; communication regions $\{\mathcal{D}_k\}_{k=i}^{j}$; previous service point FOP_{i-1}; next attractor $\mathbf{a}_{j+1}$

Output: Selected candidate sequence $\mathcal{S}_m$ and evaluation time T_m

Generate the candidate FIP set of g_i and set it as $\mathcal{L}_0$;
for $k = i$ **to** $j-1$ **do**
  Generate switching candidates in $\mathcal{D}_k \cap \mathcal{D}_{k+1}$ by the T-shaped sampling method;
  Set the generated candidates as layer $\mathcal{L}_{k-i+1}$;
Generate the candidate FOP set of g_j and set it as $\mathcal{L}_{j-i+1}$;
if *g_i is the first GN in the visiting order* **then**
  Set $\mathrm{FOP}_{i-1} \leftarrow \mathbf{q}_0$;
foreach $p \in \mathcal{L}_0$ **do**
  Set $\Phi_0(p) \leftarrow \|p - \mathrm{FOP}_{i-1}\| / v_{\max}$;
  Set $\mathrm{Pred}_0(p) \leftarrow \emptyset$;
for $r = 0$ **to** $j - i$ **do**
  Set $k = i + r$;
  foreach $q \in \mathcal{L}_{r+1}$ **do**
    Compute $\Phi_{r+1}(q)$ by selecting the predecessor $p \in \mathcal{L}_r$ with the smallest accumulated evaluation time;
    Store the corresponding predecessor as $\mathrm{Pred}_{r+1}(q)$;
Select $p^{\mathrm{end}} \in \mathcal{L}_{j-i+1}$ with the smallest value of $\Phi_{j-i+1}(p) + \|\mathbf{a}_{j+1} - p\| / v_{\max}$;
Compute T_m according to the selected p^{end};
Backtrack from p^{end} using the stored predecessors to obtain $\mathcal{S}_m$;
return $\mathcal{S}_m$ and T_m;

Since a group may contain more than two GNs, directly enumerating all possible combinations of entry points, switching points, and exit points may increase the search cost. BO therefore organizes the candidate points into layers. The first layer contains candidate FIP points of the first GN in the group, the intermediate layers contain candidate switching points generated in the overlapping regions, and the last layer contains candidate FOP points of the last GN. A dynamic programming search is then applied over these layers to find the candidate sequence with the smallest accumulated evaluation time.

After the trajectory of the group is determined, the selected FOP_j is used as the previous service point for the next planning step. When the next GN is independent, the single-node trajectory planning method in Section 5.2 is applied. When another overlapping-node group is encountered, BO is applied again. The detailed procedure of BO is summarized in Algorithm 2.

Chapter 5

Performance Evaluation

This chapter evaluates the performance of the proposed Reference-Path Attractor and Boundary-Overlap (RPA-BO) scheme for UAV-assisted data collection in wireless sensor networks. The main objective of the evaluation is to examine whether the proposed scheme can reduce the mission completion time (MCT) under different data sizes. In addition to MCT, the total flight distance is also evaluated, since the flight distance directly affects the UAV flight time and also reflects how different trajectory planning schemes guide the UAV through the GN communication regions.

The performance of the proposed RPA-BO scheme is compared with two baseline schemes. The first baseline is the Convex shortest path, which determines the shortest geometric trajectory through the GN communication regions based on the obtained GN visiting sequence. This baseline represents a geometric trajectory planning approach that mainly reduces the flight path length. To ensure a fair comparison, the same GN visiting sequence is applied to all compared schemes. The second baseline is the V-shaped scheme, which adopts a V-shaped data collection trajectory for each GN and considers the data collection process inside the communication region. The proposed RPA-BO scheme further uses reference-path attractors to provide global geometric guidance and applies the Boundary-Overlap Strategy to enable flexible service switching in overlapping GN communication regions.

5.1 Simulation Setup

The simulation area is set to 2000×2000 m^2. A total of 20 GNs are randomly and uniformly deployed in the simulation area, and a single UAV is used to collect data from all GNs. The data center is located at $(1000, 1000)$ m. The UAV departs from the data center, visits all GNs according to the generated visiting order, collects the required data from each GN, and returns to the data center after completing the mission. For each data size, 200 independent simulation runs are conducted with different random GN deployments. The reported MCT and flight distance results are the average values over these 200 runs. The required data amount of each GN is denoted by C. The data amount is set to $C \in \{1, 2, 4, 8, 16, 32\}$ MB. Table 6.1 lists the main simulation parameters used in the performance evaluation.

Three schemes are compared in the simulation. The first scheme is the Convex shortest path, which computes the shortest geometric path passing through the communication region of each GN. This scheme is used as a geometric path baseline because its main purpose is to reduce the flight distance through the GN communication regions. The second scheme is the V-shaped scheme, which adopts the V-shaped data collection trajectory. This scheme considers the trajectory inside the communication region of each GN and adjusts the UAV service trajectory according to the data amount. The third scheme is the proposed RPA-BO scheme. This scheme uses reference-path attractors to provide global geometric guidance during local trajectory planning and

TABLE 5.1: Simulation parameters.

Parameter	**Value**
Simulation area	2000×2000 m^2
Data center position	$(1000, 1000)$ m
UAV altitude	50 m
UAV maximum speed	20 m/s
GN transmit power	-40 dBm
Carrier frequency	2 GHz
Bandwidth	2 MHz
Noise power spectral density	-174 dBm/Hz
SNR threshold	2.6 dB
Path-loss exponent	1.3165
LoS additional loss	3 dB
NLoS additional loss	13 dB
LoS probability parameters	$\alpha = 11.95,\ \beta = 0.14$

applies the Boundary-Overlap Strategy to handle consecutive overlapping GN communication regions. When adjacent communication regions overlap, the UAV can select a switching point inside the overlapping region instead of leaving one communication boundary and entering another. For a fair comparison, the three schemes are evaluated under the same GN deployment and the same visiting order in each simulation run. Therefore, the differences in MCT and flight distance mainly result from the trajectory planning and data collection strategy adopted by each scheme.

5.2 Performance Comparison

Fig. 5.1 shows the initial trajectory of the UAV in the 2 km ¡ 2 km area. In this simulation scenario, 20 GNs are randomly and uniformly deployed in the area. By calculation, the communication radius (D) is approximately equal to 200 m. The UAV directly visits the center of each GN according to the visiting order. This trajectory is used to illustrate the GN distribution and the visiting order of the UAV before applying the proposed trajectory design.

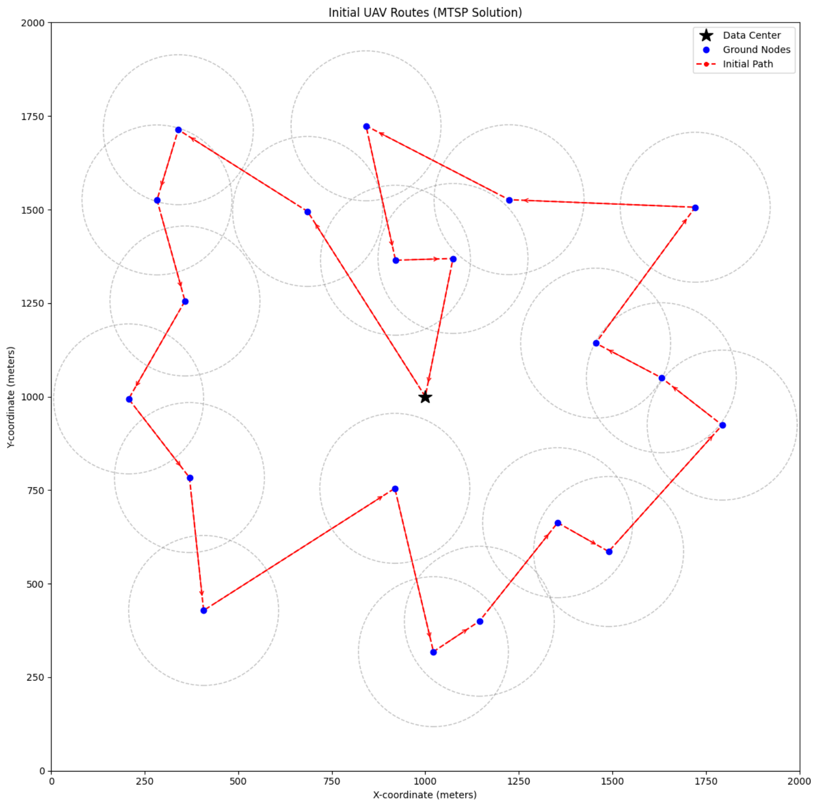

Figure 5.1: Initial trajectory of the UAV in a 2 km × 2 km area.

Fig. 5.2 compares the final trajectories of the proposed RPA-BO scheme, Convex shortest path, and V-shaped scheme under the 8 MB data size. In this case, the Convex shortest path has the shortest flight distance of 4407.78 m, but its MCT is 488.59 s. This result indicates that a shorter geometric flight distance does not necessarily lead to a shorter MCT, because the UAV may collect data near the communication boundary, where the transmission rate is relatively low. The V-shaped scheme reduces the MCT to 341.12 s by moving the UAV deeper into the GN communication region, so that the UAV can collect data at a higher transmission rate, its flight distance increases to 6766.34 m. The proposed RPA-BO scheme achieves the shortest MCT of 303.40 s with a flight distance of 6053.94 m.

This improvement mainly comes from two aspects. First, unlike schemes that restrict the entry and exit points to the boundaries of communication regions, the BO

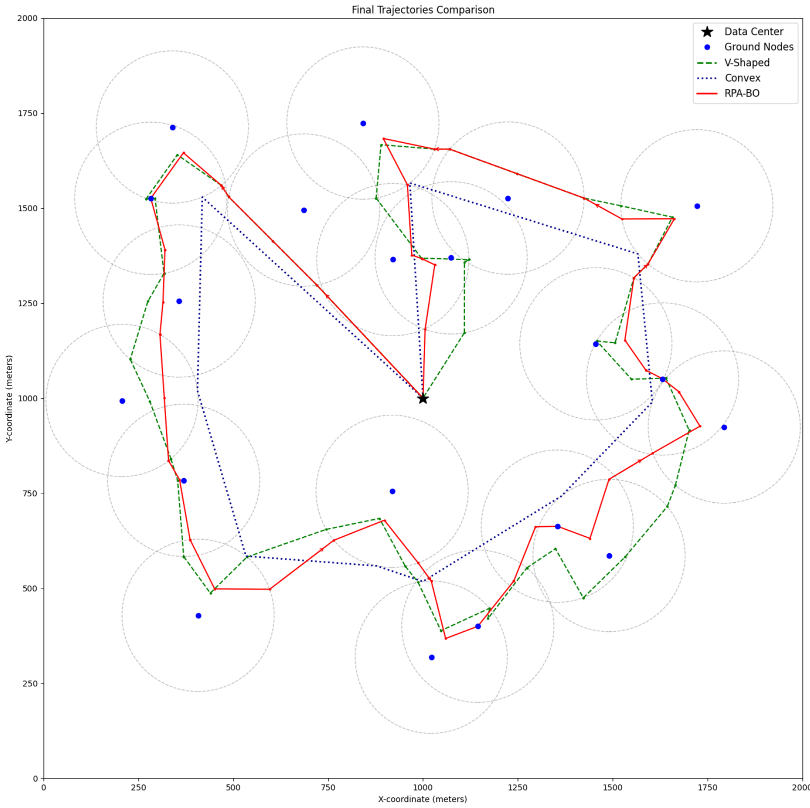

Figure 5.2: Final trajectories of compared schemes under 8 MB data size.

strategy optimizes the service switching locations within overlapping communication regions. As a result, after completing data collection from one GN, the UAV can immediately switch to serving the next GN within the overlap area, without first leaving the communication region of the current GN and then entering that of the next GN. This switching strategy reduces unnecessary flight distance and redundant movement between neighboring communication regions. Second, the RPA mechanism provides global geometric guidance during local trajectory planning, so the UAV trajectory can remain closer to the global geometric direction while still considering the data collection requirement. Therefore, the proposed RPA-BO scheme can reduce the flight distance and shorten the mission completion time.

Fig. 5.3 shows the average MCT of the three schemes under different data sizes. Each point represents the average result over 200 independent random GN deployments. The

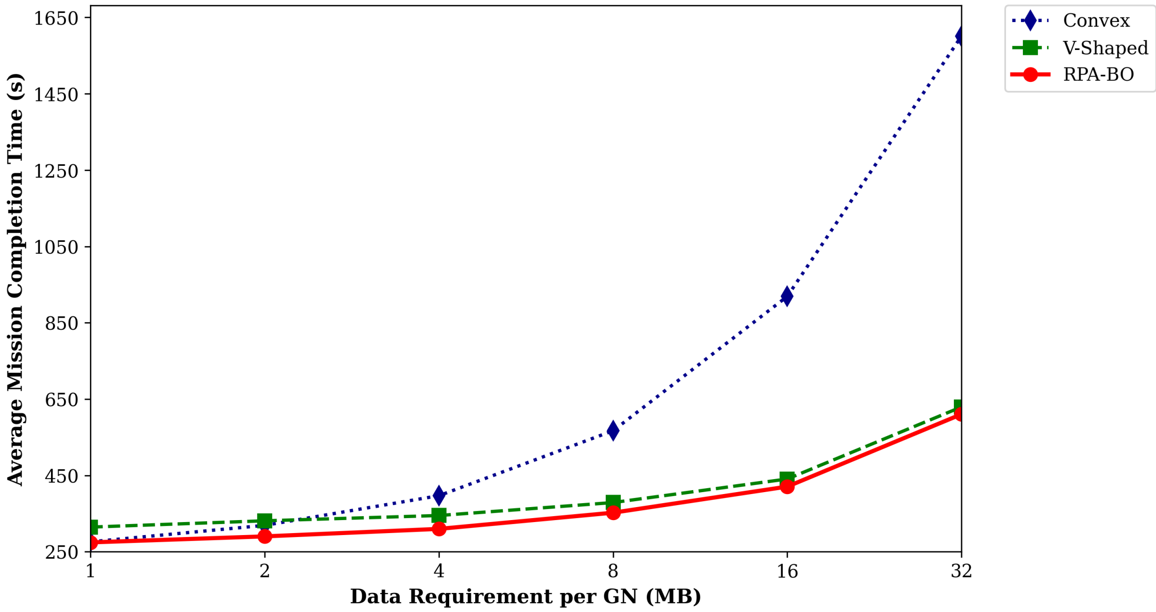

Figure 5.3: Mission completion time under different data sizes.

proposed RPA-BO scheme achieves the shortest average flight distance under all tested data sizes.

When $C = 1$ MB, the average MCT values of the Convex shortest path, V-shaped scheme, and proposed RPA-BO scheme are 276.14 s, 314.66 s, and 274.57 s, respectively. In this low-data-size case, the required data amount is small, so the flight distance is a major factor affecting the MCT. Nevertheless, the MCT of the proposed RPA-BO scheme is lower than that of the Convex shortest path because the proposed scheme does not merely reduce the flight distance, but also considers the data collection trajectory inside the GN communication regions. Even when the required data amount is small, selecting appropriate data collection trajectories and overlap switching points can reduce the time spent on service transitions and data collection, thereby leading to a shorter overall MCT.

When $C = 4$ MB, the average MCT values of the Convex shortest path, V-shaped scheme, and proposed RPA-BO scheme are 396.95 s, 345.01 s, and 309.98 s, respectively.

Compared with the Convex shortest path and the V-shaped scheme, the proposed RPA-BO scheme reduces the average MCT by 86.97 s and 35.03 s, respectively. This result shows that the proposed RPA-BO scheme can effectively reduce MCT when both flight distance and data collection time have influence on the mission.

When the data size further increases, the proposed RPA-BO scheme remains effective. When $C = 32$ MB, the average MCT values of the Convex shortest path, V-shaped scheme, and proposed RPA-BO scheme are 1600.92 s, 629.42 s, and 610.41 s, respectively. Unlike approaches that mainly focus on reducing the flight distance, the proposed scheme also improves the efficiency of the data collection process. As the required data amount increases, communication efficiency becomes increasingly important. The proposed RPA-BO scheme effectively reduces data collection time and maintains the shortest MCT among all compared schemes.

The simulation results show that the proposed RPA-BO scheme achieves the shortest average MCT under all considered data sizes. Although the Convex shortest path obtains the shortest flight distance, its MCT increases rapidly as the data size increases. This result indicates that reducing flight distance alone is insufficient when the data size increases. The V-shaped scheme reduces the data collection time by moving the UAV deeper into the GN communication region when necessary, but this also increases the flight distance. The proposed RPA-BO scheme further reduces MCT by using reference-path attractors and overlap switching points. The reference-path attractors provide global geometric guidance during local trajectory planning, while the Boundary-Overlap Strategy allows the UAV to make better use of overlapping communication regions. Therefore, the proposed RPA-BO scheme can reduce flight distance and shorten the overall MCT under different data sizes.

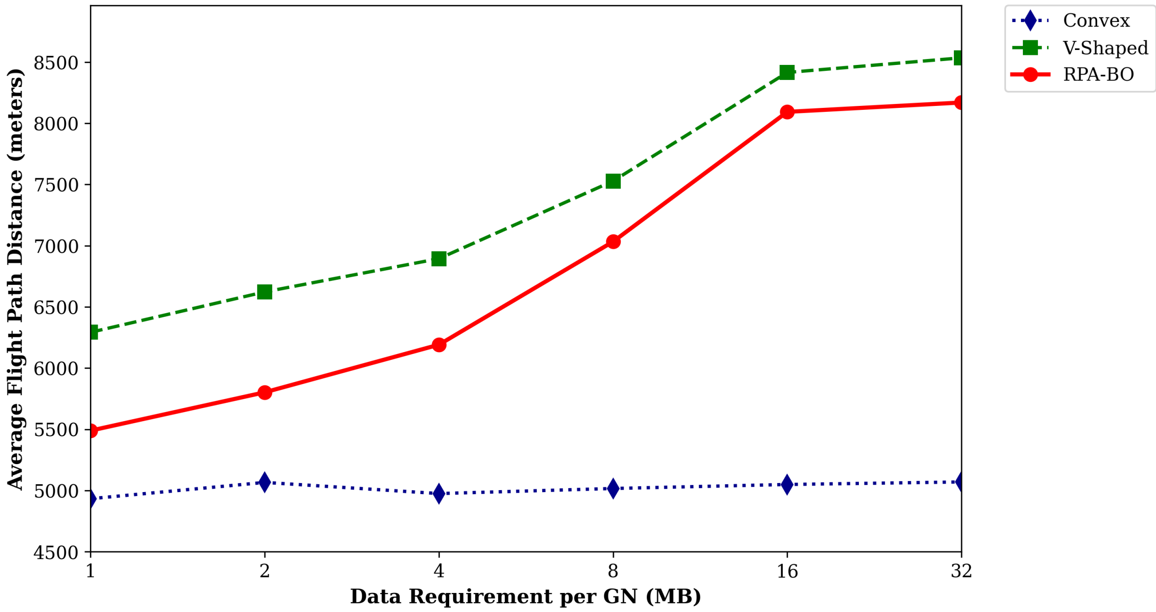

Figure 5.4: Flight distance under different data sizes.

Fig. 5.4 shows the average flight distance of the three schemes under different data sizes. Each point represents the average result over 200 independent random GN deployments. As shown in Fig. 8, the shortest flight distance does not correspond to the shortest MCT, especially when the required data amount becomes large. This result confirms that flight distance and data collection time should be considered together.

The V-shaped scheme has a longer flight distance because the UAV may move deeper into the GN communication regions to improve the transmission rate. In contrast, the proposed RPA-BO scheme achieves a shorter flight distance than the V-shaped scheme under all tested data sizes. When $C = 4$ MB, the average flight distances of the V-shaped scheme and the proposed RPA-BO scheme are 6894.67 m and 6191.67 m, respectively. The proposed RPA-BO scheme reduces the average flight distance by 703.00 m while also reducing the MCT, as shown in Fig. 8. When $C = 32$ MB, the average flight distances of the V-shaped scheme and the proposed RPA-BO scheme are 8534.91 m and 8170.13 m, respectively. The proposed RPA-BO scheme still maintains

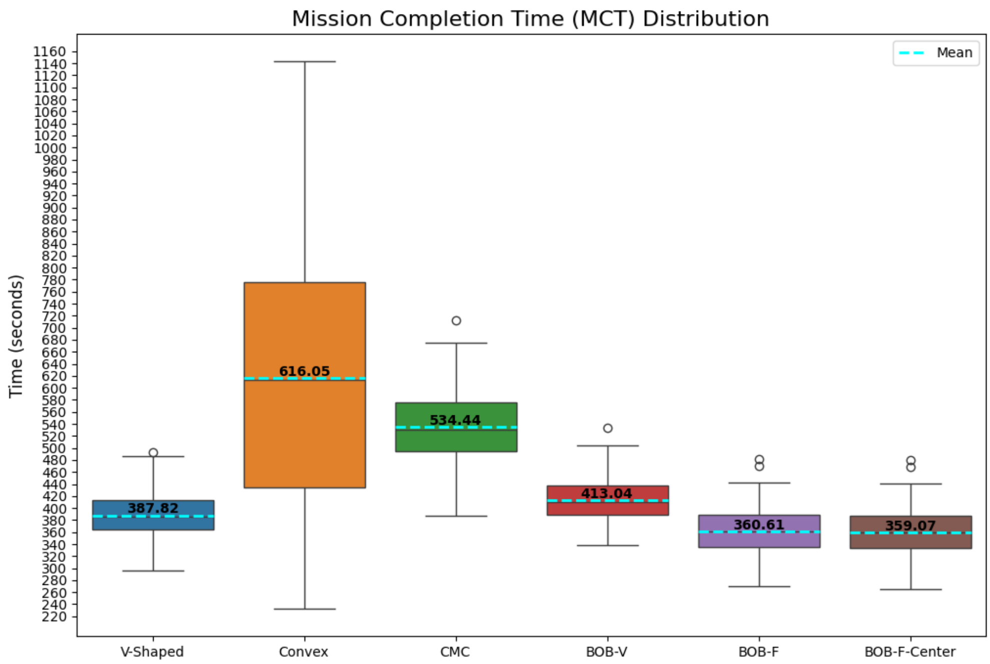

Figure 5.5: MCT distribution under heterogeneous data amounts.

a shorter flight distance. This improvement is mainly attributed to the reference-path attractors and overlap switching points. The reference-path attractors guide the local trajectory planning toward the global geometric direction, while the overlap switching points allow the UAV to switch services more efficiently in overlapping communication regions. This result indicates that the proposed scheme can reduce the flight distance while maintaining the benefits of efficient data collection.

To further evaluate the proposed scheme under heterogeneous data requirements, an additional experiment is conducted in which the required data amount of each GN is randomly generated from 1 MB to 32 MB. Fig. 5.5 shows the MCT distribution of the compared schemes under heterogeneous data amounts. The proposed RPA-BO scheme achieves the lowest average MCT among the compared schemes. The average MCT values of the V-shaped scheme, Convex shortest path, and proposed RPA-BO scheme are 387.82 s, 616.05 s, and 360.61 s, respectively. Compared with the V-shaped scheme, the proposed RPA-BO scheme reduces the average MCT by 27.21 s, corresponding to a 7.02% reduction. This result indicates that the proposed scheme remains effective when different GNs have different data requirements.

The simulation results demonstrate that reducing the flight distance alone is insufficient to reduce the MCT in UAV-assisted WSN data collection. The MCT is jointly affected by the UAV flight distance, the data collection trajectory within GN communication regions. By considering these factors together, the proposed RPA-BO scheme achieves the shortest MCT among the compared schemes under the tested data amounts. These results confirm that integrating global geometric guidance, local data collection trajectory design, and service switching in overlapping communication regions is effective in reducing the overall MCT.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This thesis studied trajectory planning for mission completion time reduction in UAV-assisted data collection in WSNs. The mission completion time was affected by both the flight time between GNs and the data collection time within each GN communication region. A shortest geometric path alone could reduce the flight distance, but such a path could also keep the UAV near the boundary of communication regions and increase the data collection time. Therefore, this thesis considered the coupling between the flight trajectory and the data collection trajectory.

This thesis proposed a trajectory planning method that combines the Reference-Path Attractor (RPA) mechanism and the Boundary-Overlap Strategy (BO). The RPA mechanism used a shortest geometric reference path to provide global directional guidance for local trajectory adjustment. For single-node service, the UAV adjusted its service trajectory inside each communication region according to the required data amount and the communication condition. This adjustment determined how far the UAV moved into the communication region and whether additional hovering time was required. For consecutive GNs with overlapping communication regions, BO allowed flexible service switching inside the overlap region and searched for a shorter continuous service trajectory. Simulation results showed that the proposed method reduced the overall MCT compared with the benchmark methods under different data requirements. The results also showed that the proposed method maintained a shorter flight distance while reducing the data collection time, which confirms the effectiveness of jointly considering flight trajectory, data collection trajectory, and overlap-aware service switching.

6.2 Future Work

Several directions can be further studied in future work. First, the current study assumes a fixed UAV altitude and a fixed flight speed except during hovering. Future studies can include altitude and speed control to further reduce the MCT under practical UAV motion constraints. Second, this thesis focuses on a single-UAV data collection task. Extending the proposed method to multi-UAV scenarios can further improve mission efficiency in large-scale WSNs. Finally, the visiting order and the local trajectory planning process can be studied jointly to further improve the overall data collection performance.

References

[1] Z. Qin, A. Li, C. Dong, H. Dai, and Z. Xu, "Completion time minimization for Multi-UAV information collection via trajectory planning," *Sensors*, vol. 19, no. 18, p. Art. no. 4032, Sept. 2019.

[2] L. Zhang, C. He, Y. Peng, Z. Liu, and X. Zhu, "Multi-UAV data collection and path planning method for large-scale terminal access," *Sensors*, vol. 23, no. 20, p. Art. no. 8601, Oct. 2023.

[3] K. Xiang and Y. He, "Uav-assisted MEC system considering UAV trajectory and task offloading strategy," in *Proceedings of 2023 IEEE International Conference on Communications (ICC): Green Communication Systems and Networks Symposium*, May 2023, pp. 4677–4682.

[4] W. Meng, X. Zhang, L. Zhou, H. Guo, and X. Hu, "Advances in UAV path planning: A comprehensive review of methods, challenges, and future directions," *Drones*, vol. 9, no. 5, p. Art. no. 376, May 2025.

[5] Z. Wei, M. Zhu, N. Zhang, L. Wang, Y. Zou, Z. Meng, H. Wu, and Z. Feng, "UAV-assisted data collection for internet of things: A survey," *IEEE Internet of Things Journal*, vol. 9, no. 17, pp. 15 460–15 479, Sept. 2022.

[6] Y. Zeng, Q. Wu, and R. Zhang, "Accessing from the sky: A tutorial on UAV communications for 5G and beyond," *Proceedings of the IEEE*, vol. 107, no. 12, pp. 2327–2375, Dec. 2019.

[7] A. Al-Hourani, S. Kandeepan, and S. Lardner, "Optimal LAP altitude for maximum coverage," *IEEE Wireless Communications Letters*, vol. 3, no. 6, pp. 569–572, Dec. 2014.

[8] Y. Zeng, J. Xu, and R. Zhang, "Energy minimization for wireless communication with rotary-wing UAV," *IEEE Transactions on Wireless Communications*, vol. 18, no. 4, pp. 2329–2345, Apr. 2019.

[9] Q. Wu, Y. Zeng, and R. Zhang, "Joint trajectory and communication design for Multi-UAV enabled wireless networks," *IEEE Transactions on Wireless Communications*, vol. 17, no. 3, pp. 2109–2121, Mar. 2018.

[10] S. Zhang, W. Liu, and N. Ansari, "Completion time minimization for data collection in a UAV-enabled IoT network: A deep reinforcement learning approach," *IEEE Transactions on Vehicular Technology*, vol. 72, no. 11, pp. 14 734–14 745, Nov. 2023.

[11] S. Shen, K. Yang, K. Wang, G. Zhang, and H. Mei, "Number and operation time minimization for Multi-UAV-enabled data collection system with time windows," *IEEE Internet of Things Journal*, vol. 9, no. 12, pp. 10 149–10 161, June 2022.

[12] M. Li, S. He, and H. Li, "Minimizing mission completion time of UAVs by jointly optimizing the flight and data collection trajectory in UAV-enabled WSNs," *IEEE Internet of Things Journal*, vol. 9, no. 15, pp. 13 498–13 510, Aug. 2022.

[13] Y. Li, W. Liang, W. Xu, X. Jia, Y. Xu, and H. Kan, "Data collection maximization in IoT-sensor networks via an energy-constrained UAV," *IEEE Transactions on Mobile Computing*, vol. 22, no. 1, pp. 159–175, Jan. 2023.

[14] T. S. Alemayehu and J.-H. Kim, "Efficient nearest neighbor heuristic TSP algorithms for reducing data acquisition latency of UAV relay WSN," *Wireless Personal Communications*, vol. 95, no. 3, pp. 3271–3285, Feb. 2017.

[15] J. Gong, T.-H. Chang, C. Shen, and X. Chen, "Flight time minimization of UAV for data collection over wireless sensor networks," *IEEE Journal on Selected Areas in Communications*, vol. 36, no. 9, pp. 1942–1954, Sept. 2018.

[16] Q. Fu, R. Jia, F. Lyu, F. Lin, Z. Zheng, and M. Li, "Collection point matters in time–energy tradeoff for UAV-enabled data collection of IoT devices," *IEEE Internet of Things Journal*, vol. 11, no. 19, pp. 31 492–31 506, Oct. 2024.

[17] Y. Wang, M. Chen, C. Pan, K. Wang, and Y. Pan, "Joint optimization of UAV trajectory and sensor uploading powers for UAV-assisted data collection in wireless sensor networks," *IEEE Internet of Things Journal*, vol. 9, no. 13, pp. 11 214–11 227, July 2022.

[18] C. Luo, W. Chen, D. Li, Y. Wang, H. Du, L. Wu, and W. Wu, "Optimizing flight trajectory of UAV for efficient data collection in wireless sensor networks," *Theoretical Computer Science*, vol. 853, pp. 25–42, Jan. 2021.

[19] Y. Wang, Z. Hu, X. Wen, Z. Lu, and J. Miao, "Minimizing data collection time with collaborative UAVs in wireless sensor networks," *IEEE Access*, vol. 8, pp. 98 659–98 669, May 2020.

[20] S. He, H. Qin, and K. Wang, "V-shaped trajectory planning of search and rescue UAV based on target detection," in *Proceedings of 2024 6th International Conference on Artificial Intelligence and Computer Applications (ICAICA)*, Nov. 2024, pp. 24–29.

[21] J. Li, H. Zhao, H. Wang, F. Gu, J. Wei, H. Yin, and B. Ren, "Joint optimization on trajectory, altitude, velocity, and link scheduling for minimum mission time in UAV-aided data collection," *IEEE Internet of Things Journal*, vol. 7, no. 2, pp. 1464–1475, Feb. 2020.

[22] X. Yuan, Y. Hu, J. Zhang, and A. Schmeink, "Joint user scheduling and uav trajectory design on completion time minimization for uav-aided data collection," *IEEE Transactions on Wireless Communications*, vol. 22, no. 6, pp. 3884–3897, Jun. 2023.